REGENERATIVE BRAKING SYSTEM

BY

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Abstract

A Regenerative Braking System (RBS) is an energy recovery technology used in electric and hybrid vehicles. During braking, instead of wasting the vehicle's kinetic energy as heat through conventional friction brakes, the system converts this energy into electrical energy using an electric motor operating as a generator. The generated electrical energy is stored in the battery and can be reused to drive the vehicle. This improves energy efficiency, increases driving range, reduces brake wear, and lowers energy consumption. Regenerative braking is therefore an important technology for developing efficient and environmentally friendly electric vehicles.

Regenerative Braking System

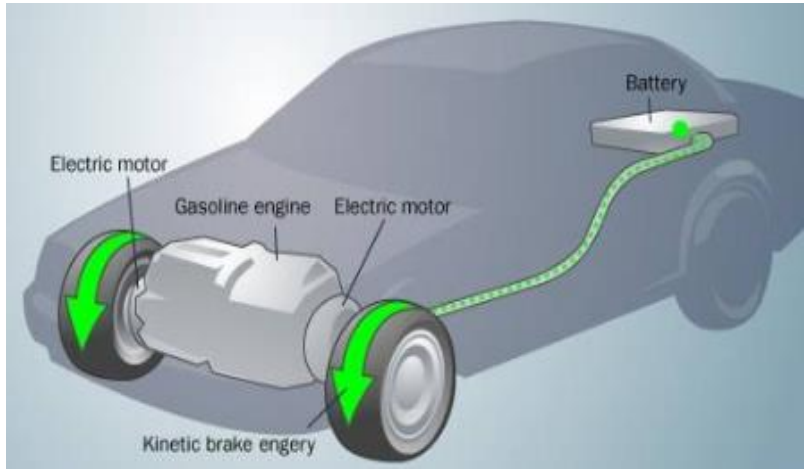
An efficient braking technology used in electric and hybrid vehicles

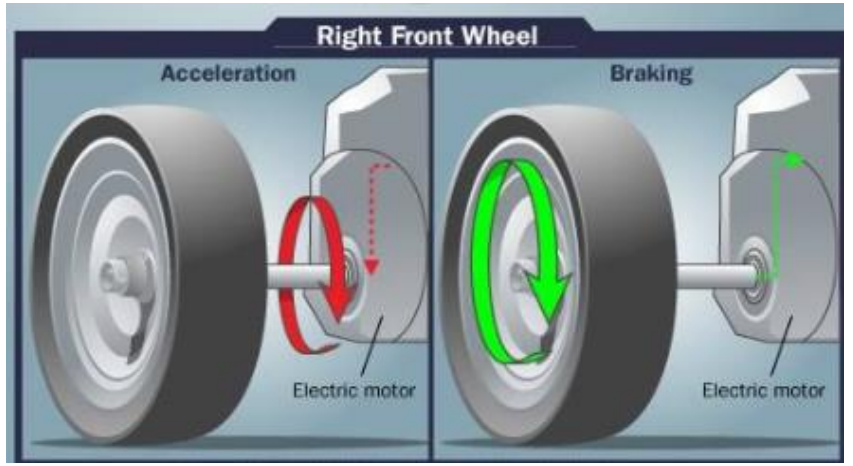
Converts vehicle kinetic energy into electrical energy

Prepared for: ECE / Automobile Engineering

1. Introduction

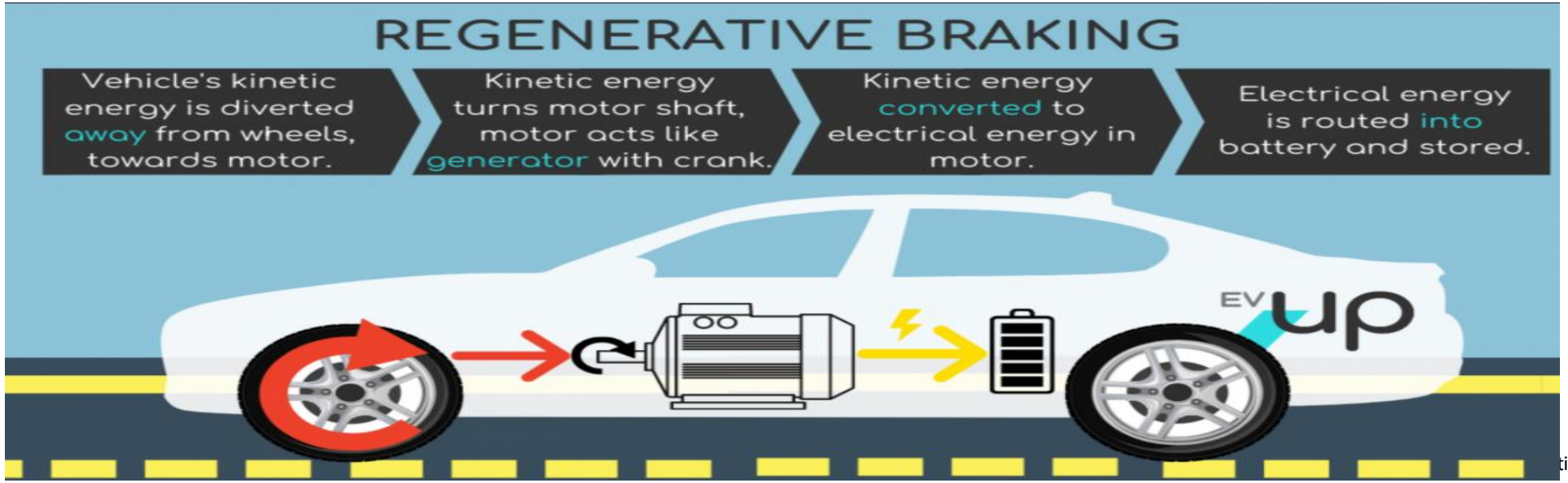
- Regenerative braking recovers energy normally lost as heat during braking.
- The electric motor operates as a generator when the vehicle slows down.
- The generated electrical energy is stored in the battery for later use.
- It improves energy efficiency and reduces brake wear.





2. Working Principle

- During acceleration, the motor uses battery power to drive the wheels.
- During braking, the motor changes into generator mode.
- The rotating wheels drive the motor and produce electrical energy.
- A power electronic controller regulates the generated current.
- The recovered energy is sent to the battery.

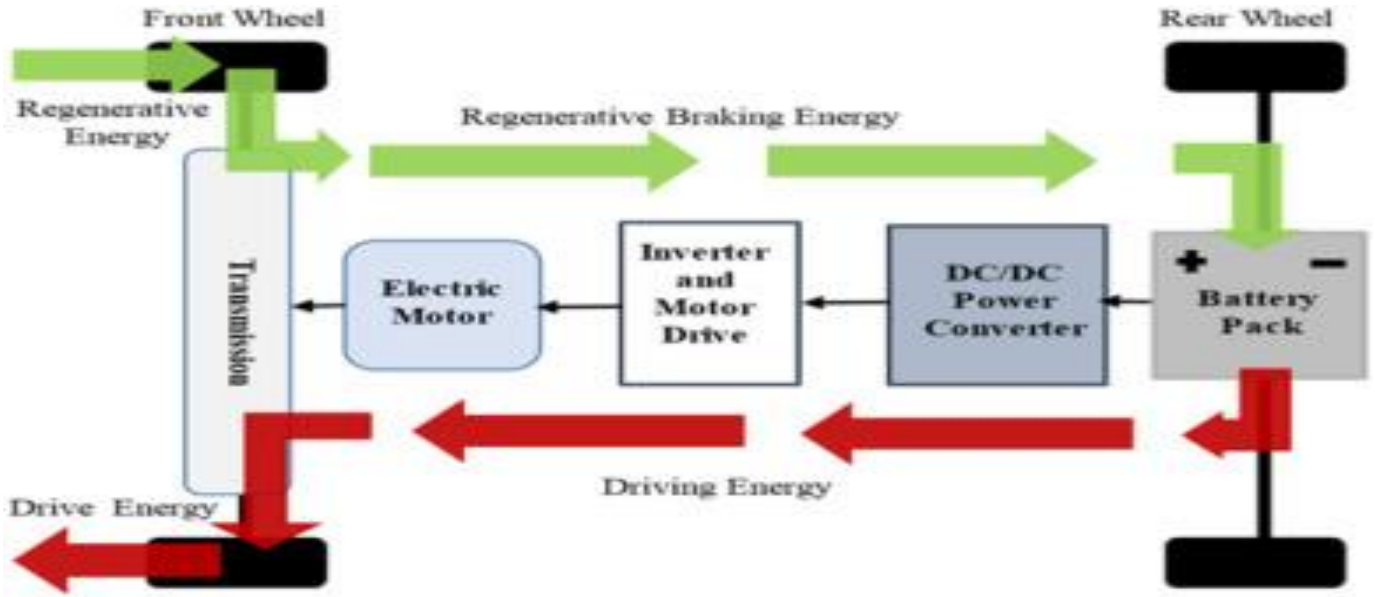


3. Block Diagram

Energy flow during regenerative braking:

Wheels → Electric Motor/Generator → Power Converter → Battery

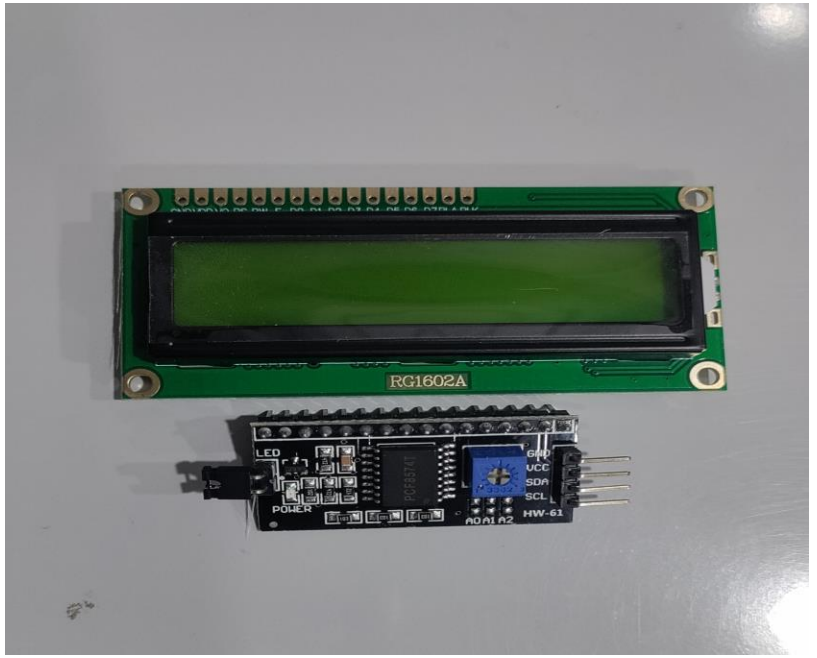
- Brake Controller coordinates regenerative and mechanical braking.
- Mechanical brakes provide additional braking when required.



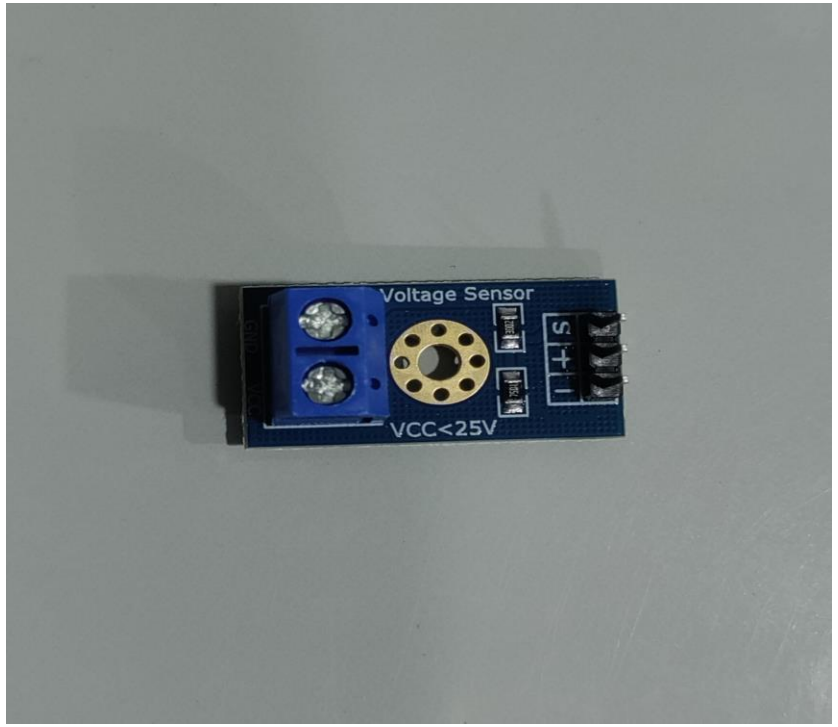
4. Main Components

- Electric motor/generator – converts mechanical energy to electrical energy.
- Battery pack – stores recovered electrical energy.
- Inverter / DC-DC converter – controls voltage, current and power flow.
- Brake controller – manages braking force and energy recovery.
- Mechanical brakes – provide supplementary braking.


ARDUINO
UNO


RG1602A
LED
POWER
GND
VCC
SDA
SCL
HW-61


Hi-Watt
6F22 9V BATTERY
GENERAL PURPOSE

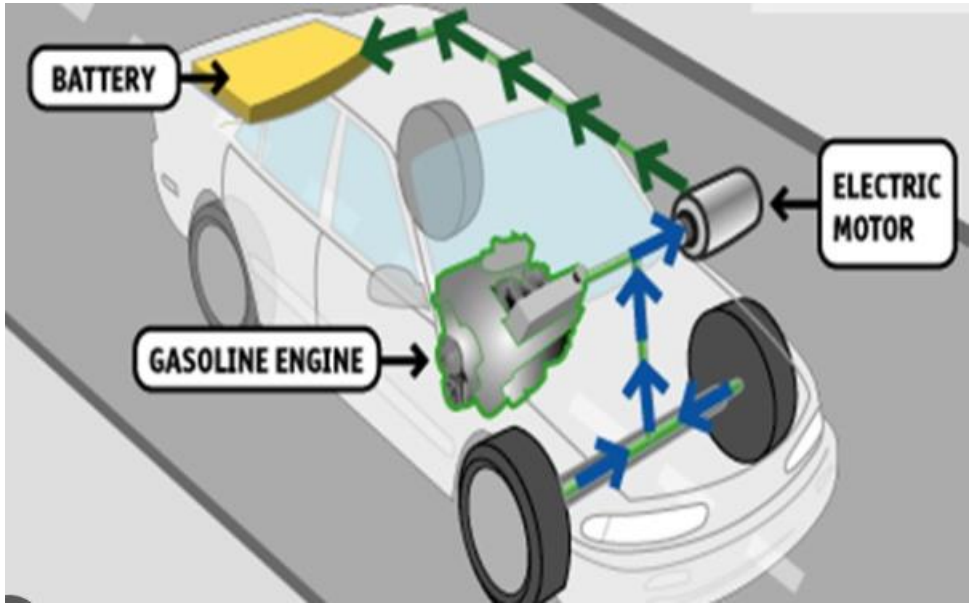

Voltage Sensor
VCC<25V





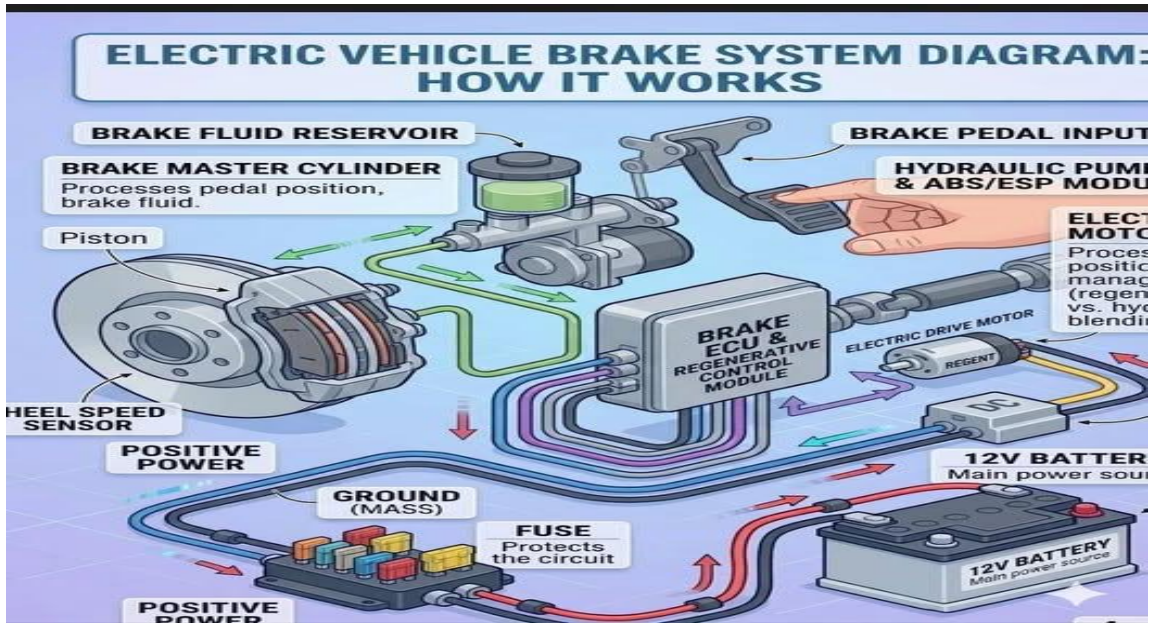
5. Advantages

- Recovers energy that would otherwise be wasted as heat.
- Improves vehicle energy efficiency and driving range.
- Reduces wear of conventional brake pads and discs.
- Produces less heat during repeated braking.
- Useful in electric, hybrid and regenerative railway systems.



6. Limitations & Applications

- Energy recovery is limited when the battery is fully charged.
- Regenerative braking is less effective at very low speeds.
- Requires additional power electronics and control systems.
- Applications: EVs, hybrid cars, electric buses, trains and some industrial vehicles.
- Mechanical braking remains necessary for emergency and low-speed braking.



7. Conclusion

- Regenerative braking converts braking energy into useful electrical energy.
- It is an important technology for efficient electric and hybrid transportation.
- Combining regenerative and friction braking provides effective and safe braking.
- Future improvements in batteries, motors and control systems can increase energy recovery.

THANK YOU 